Archival Preprint Note

This is an archival version of theoretical work completed 2020–2022 and included as Chapter 2 of the PhD dissertation

"Essays in Market Microstructure: On the Information Content of Quotes and Continuous Auctions in Order Driven Markets"

by Sudhanshu Pani (NMIMS University, 2022), available at:

https://shodhganga.inflibnet.ac.in/handle/10603/466909

A revised and extended framework, including empirical tests in decentralized markets (Uniswap v3 liquidity pools), is in preparation.

Markets as Trading Systems: An Intraday Perspective on Limit Order Books without Market Makers

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February 2026

(Archival preprint version 2020–2022)

Abstract

This article investigates a limit order market in high resolution. The market is presented as a trading system. It offers an improved alternative to the current paradigm of the mechanism view along with the game theoretic approach to examine markets. An intraday trading model is described where trading is motivated by a need to change the state of the asset held. The model is characterised by a time horizon of a single trading day, absence of market makers and trading in disequilibrium. The trading system controls the trading intensity to control the wealth traded. The role of markets as places of exchange is emphasised along its role in price discovery and information processing.

JEL CODES: J12, J14

KEYWORDS: Systems view, Trading system, Information content of orders; Information Aggregation; High Resolution; Limit Order Markets; Intraday Trading;

1 Introduction

Order-driven markets have ensured that information that used to earlier reach only the dealer or market maker is now available in the data recorded by the exchanges. O'Hara (2015) noted that the orders were the basic unit of information and suggested the microstructure toolbox to be enhanced in order to investigate this. The event times recorded in any such data from modern markets is likely to be of the order of sub-second time intervals. Hence, Hasbrouck (2021) and Pani (2021) refer to such examination as high resolution market microstructure. The theoretical framework that can improve our understanding of the markets when viewed in high resolution is the subject of this paper.

1.1 The trading system view versus the mechanism view of the market

We consider the market, in high resolution, as a trading system. This framework contrasts in several ways with the current prevalent mechanism view of the market. Wilson (2021) presents an historical account of the evolution of the study of incompletely competitive markets. To start, the key observables in the market were the prices and quantities. A fixed point in prices and quantities characterised the equilibrium. Then, the game theoretic approach, tried to incorporate certain unobservables, specifically the information, actions and trading strategies of the agents. This approach relied on Hurwicz (1973) view of a market as a mechanism. The equilibrium is characterised as a fixed point in agents strategies.

However, when viewed in high resolution in the context of markets today, the current framework that has evolved from the mechanism view together with the game theoretic approach is handicapped. Consider the excellent treatiese on mechanism design, Myerson (1989) and mechanism and economic theory, Potts (2001). The concept of mechanism can be characterised as being fluidic since the eighteenth century. The modern interpretation associates the market as an information processing mechanism. Market microstructure

additionally considers several frictions present during trading of assets in the markets. 'Information incorporation in security prices' , 'Private information incorporated into market prices' has been an important and recurrent theme in microstructure models.

For example, in Kyle (1985), an insider (a large trader with private information), noise traders and the market maker are engaged in sequential auctions leading to a sequential equilibrium. If trading takes place frequently this can converge to a simple model of continuous trading. Information is incorporated into the prices. In Glosten and Milgrom (1985), the idea that bid-ask spread has asymmetric information as one of the source, is used. The spread is required to take care of losses due to adverse selection. The spread is modeled to ascertain the information flow in the market. Back (1992) extends the Kyle (1985) model to continuous time.

Similarly, in Collin-Dufresne and Fos (2016), Kyle's model is generalized to allow the volatility of noise trading to change stochastically over time. This model includes market makers, a notion that has spilled on from the era of quote driven markets. While some participants may indulge in market making, designated market makers do not exist. O'Hara (2015) calls such constructs (the presence of market makers or invisible auctioneers invoked to enforce equilibrium) as 'fictional'. The need for the presence of a market maker or similar entity is an essential requirement in mechanism design. (Myerson (1989)). Here the objective is to ascertain what combination of agent strategies can lead to pre-set end goals. Our view is that in high resolution as the effects of market microstructure dominate, the absence of effective ways of communication among agents or any central market maker handicaps the game theoretic approach based on the mechanism view.

Mechanism

Market mechanism is a rule system for communicating price information Hurwicz (1973). This concept has its origins in physics and information theory. The mechanisms in both

these domains are defined in terms of closed system. However, when it comes to serveral systems involving human agents, the mechanisms deal with open systems. Such open ended mechanisms are called as processes (Potts (2001)). Our hypothesis is that if the need for communication in sub-second intervals is lowered an adequate mechanism can be designed.

System

Markets have in the past been viewed as systems. The notion of a system as an association of parts into a functional form has withstood any change over time. Hence, in the mechanism view too the market is in effect a system. Unlike the mechanism view, in the trading system view, the system view dominates over individual agent actions. All agents have a similar key goal, that of effecting a change of state in their holdings. This in effect becomes the goal of the trading system, and the equilibrium in the system view is characterised in the optimal rate of transactions (or state transformation). This in effect reduces the dimensionality of the problem. Additionally, the optimal rate of transaction shall respond if the agent objectives for state transformation of the asset as per their planned schedules is not met. Note this is a crucial difference between the two approaches.

Optimal trading strategies versus Optimal trading rate

In the mechanism view we model to evaluate optimal trading strategies between agents to achieve a given equilibrium. In the trading system view neither do we seek to achieve a target optimal, nor do we seek to achieve optimal agent trading behaviour. Agent strategies are not revealed due to lack of time or ways to communicate. However when a good number of agents change their execution strategies it impacts the system properties beyond a threshold. The system may move from one local equilibrium to another where the equilibrium is an optimal trading rate.

We contribute to the literature by presenting a new framework to examine high resolution market microstructure. This is achieved in the following ways. Most order driven equity markets do not designate official market markers or dealers for the securities. First, this article examines the information content of quotes without the presence of a market maker in the model. A model without market maker was also used in Vayanos and others (1993), Vayanos (1999)and Kyle et al. (2018). Baruch and Glosten (2019) model limit order book markets under imperfect competition, quote cancellations and tail expectations that arise in such markets due to the certain rigidities in such markets. Second, we relax the market clearing condition (based on the tatonnement process "auction as a search" Pani (2020)). This helps us to work with locally optimal solutions and not look for the global equilibrium. The result is that we are able to restrict the analysis to an Intraday model. Third, we use both price and quantities as we base our analysis on the wealth in the limit order book. Fourth, we conjecture in our model, that there exist two information signals in the market tracking two time scales. Fifth, information flows through the shocks into a conjectured 'Information content structure' which resembles a collective memory of the market participants. Here 'collective market memory' refers to the collective observation and inference of the market players about the price and trading dynamics in the market.

The rest of the article is organised as follows: Section 2 introduces the set up used in this article and the notions incorporated in the model. The intraday trading model is discussed in Section 3. Section 4 is a discussion on microstructure and trading effects. Section 5 presents the concluding remarks.

2 Intraday Trading: Introduction, The Set Up, Assumptions and Notions

In this section we consider an intraday trading model. This model attempts to present an analysis of some aspects of limit order market that is closer to reality. We do not

specify the behaviour of individual agents and we only rely on the aggregate effects of such behaviour observed in the trading system. Our model describes any typical trading day (unlike for example Vayanos (1999) or Kyle et al. (2018) where trading may take place over several days) and hence is a trading micro model. Our notion of equilibrium is that transitory equilibrium may occur during the day, but the traders may not be aware about it. While we descrie a single trading day, the system is an open system, i.e the asset is traded both prior and after the given trading day under analysis. The model does not depend on equilibrium and market clearing for price discovery as it assumes price could also be discovered as a search in auctions (Pani (2020)). Early trade or a pre-auction is an opportunity to strive for some equilibrium price. If quantity trades are not executed in the auction convective processes set in that attempt to reduce volume biases. These processes could be driven by new quotes coming in from the latent order book or quote modifications.

2.1 The Set Up

We consider a market for a single risky asset. The market is organized as a pure limit order book deploying a continuous auction mechanism with the usual price-time priority. The asset price Y is a random variable, $0 < Y < \infty, Y \in R$. The total number of securities of the asset be n. The public signal of the asset price is given by y_p which is considered as the asset price by a large number of active market participants and is conditioned on the available public information . Since we are interested in the private information, we assume that the public information is available with all market participants and is interpreted in the same way resulting into the value y_p for each participant. Any news that is public information may get interpreted differently by some traders, this updates their private information. Fig. 1 illustrates the set up.

Traders

There are two kinds of traders in the market; Strategic traders ST, $ST \in N, 1 \leq ST \leq \infty$, who during the trading day come to the market with private information e and non-strategic traders NST,$NST \in N, NST_{min} \leq NST \leq \infty$ who do not have any private information. Asymmetric Information leads to strategic behaviour on price dimension[1] All traders can choose to deploy a limit or market order to execute their trades. There however necessarily exists a minimum number of non-strategic traders M_{min} who deploy market orders to avoid the no-trade equilibrium due to non-availability of market orders. We conceive this by considering that a part of the non-strategic traders continuously trade in the market by alternatively placing a limit order and market order. We call this subset as the non-strategic noise traders. The number of strategic buyers ST_B ,$1 < ST_B \leq \infty$ and strategic sellers ST_S ,$1 < ST_S \leq n$. There will always exist atleast one strategic buyer and seller. The maximum number of strategic sellers is restricted to the number of securities. $s_B Y, t$ and $s_S Y, t$ are the pdf (probability density function) of strategic buyers and sellers respectively, where Y is price and t is time . So $m_B y_p, t \;\; m_B y_p, t \;\; s_B Y, t \;\; s_S Y, t = 1$ where $m_B$and m_S are the pdf of non-strategic traders conditioned on the public signal. This shows that the presence of strategic traders depends on the price and time. Strategic traders come into the market only when price is at Y or better. Y is the one-sided boundary for strategic traders. Only a part of the Total or latent order book reveals as limit orders in the visible order book. The non-strategic traders are present in the market conditioned on the public signal. Such traders do not have any private information. Unlike classical noise traders, the non-strategic noise traders try to make a non-negative profit during a trading day.

[1]The use of nomenclature strategic and non-strategic emanates from the broad definition of Information in today's markets. Every trader may possess information of some kind. The trader has several choices to utilise the information in trading. The choices range from price, order types, quantities, timing, sequence, algorithms etc. To simplify, we ask: Does the trader use the information in the price dimension? If yes, we view this as a strategic behaviour in this article. More commonly used terminology for trader types in microstructer models is informed traders versus non-informed traders (O'Hara (2015),Gould et al. (2013)). Models without market makers have depended on strategic motives: Strategic Informed Traders in Kyle et al. (2018), Strategic agents in Vayanos (1999) are large traders and only traders in the model. Noise Traders (as in Kyle (1985)) is smaller subset of non strategic traders.

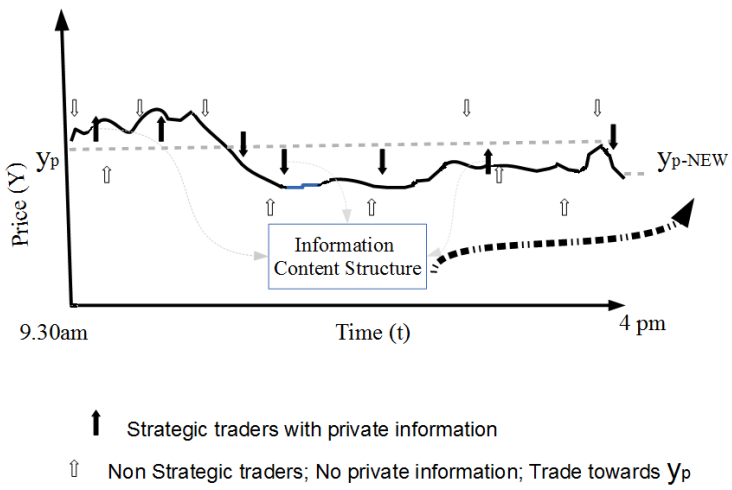

Figure 1: The Set up illustrating the presence of strategic traders and non-strategic traders. The public signal at the start of the day is y_p that gets revised at the end of the day. Strategic traders quote with reference to the price and non-strategic traders refer the last traded price and move in the direction of the public signal y_p. The market records the volume-weighted price shock of all traders in the ICS.

A trader can communicate outside or inside the market. We assume that communication outside the market implies only an update into the public signal and private signals of other participants is not impacted. Thus information flow constraints do not exist. If any trader communicates outside the market, it reaches every other trader and they do not update their private signal as the information is with all traders. While y_p can vary with time, we assume it does not change during the course of a single trading day which is our unit of analysis. An instance where y_p can receive an update is when there is a general price discovery auction such as the pretrade-auction implemented by several exchanges. We assume no such price discovery auction is implemented within the trading day. The existing y_p^0 is an update arrived after the previous day's trading or the pre-auction. The new update after the current trading day would be y_p^1 after it has incorporated the private information. In this way any news that arrives during the trading day is incorporated into the public signal through the private information or at the end of the trading day.

The reserve price for any trader is given in (1), where the private information based signal of ith trader is e_i. e_i is zero for a non-strategic trader.

$$Y_i = y_p \;\; e_i \tag{1}$$

While the preference of an agent may be to execute his total trade at a price Y_i, he may not be able to execute it at the price in a competitive auction. The agent has his own demand and supply schedule around Y_i. If he gets a price superior to Y_i he takes it. He finally comes into the market through a series of orders at different price and quantity schedules. Note that the total of these quantities and the prices would get revised through the series of sequential auctions depending on the evolving winning trades. The trader's bids or offers are, however, conditioned by his reserve price. The transaction price arrived at from the auction of a partial number of the available quantity of an asset can only be partially informative. (Refer Appendix A for the proof)

In summary, when only strategic traders have private information and hence may choose to trade at specific prices and non-strategic traders do not have private information and trade around the public signal, if the transaction price is fully informative it would replace the public signal. The public signal is assumed to be updated at the end of the trading day, but the transaction price would update the public signal with every transaction. Every asset owner is better off participating in the auction and liquidating some of his asset at each auction. Asset owners cannot choose not to participate and trade. Only strategic traders would be available as buyers in several price regions and hence we would have more no-trade equilibrium.

If on the other hand the total quantity of the asset n or a number which is very large, is bid for and transacted in the trading day, the auction price can be fully informative. Since the quantity transacted is large, limited private information can come in through any separate bid or offer. Agreement between the buyer and seller at the transaction price implies all the available public and private information is incorporated.

2.2 Notions and assumptions incorporated in the model

We now introduce a few notions and assumptions that are incorporated into the model.

2.2.1 Trading is smooth intraday but not between dates

Large traders or investors cannot execute their mandates in a single trading day. Hence, our contention is that while trading can be smooth within a trading day, it cannot be smooth across trading dates. This is due to what we call a 'time-date' issue. The time gap between two events in the market (order book) is of the order of millisecond but the time gap between two trading days is several orders of magnitude larger. Within this period trading remains suspended but news flow, analysis and realignment remains continuous. This large time gap, in which there is no trading, can enable new information to flow in and exisiting market and trading information to be assimilated. If large traders factor in some amount of price impact in their quoting behaviour it may be construed that they possess private information.

2.2.2 Losses

Some agent has to absorb the loses in a trading system. A loss in a trading system is notional, in the sense that every transaction has two parts. Which leg gains or loses depends on the deviation of the transaction price from their expected prices. Who absorbs the losses in the absence of market makers is critical. Trading models with a market maker are usually set up in such a way that either noise traders or uninformed traders absorb losses. In the event that the market maker picks losses due to adverse selection, he may have the opportunity to recoup losses from the uninformed traders. We look at loss from the perspective of an adverse change in the price of quote from planned behaviour.

We expect strategic traders to bear most of such losses and other transaction costs in the trading system. This is because they would like to actively manage the risk of non-execution or the time value of money. As a corollary, it implies that price resets to increase the trading rate can be expected from strategic traders. In a sense the intuition is that strategic traders try to cut the impact cost during trades. Non strategic noise

traders are expected to be wealth neutral in this model. Only then can they provide the bridge to the non-synchronous arrival of buyers and sellers in the model in the absence of market marker.

2.2.3 The System view and Utility functions

Imagine we build a system with artificial intelligence that has access to the needs and motivations of all traders. Such a system can optimise the outcome by setting the appropriate rate of transaction. The utility represents the utility of such a system. Individual agents have their own utility funtions. However, the trading system is influenced by the system level dynamics and utility, which is simply the aggregate of the individual behaviour. Bone (1998) showed that if we assume the individual utilities of traders are CARA (constant absolute risk-aversion), then the collective choices that emerges conforms to EU (expected utility theory). A source of error emerges from the fact that traders have access only to the visible order book and not the latent order book that has to be inferred. Trader quotes arrive at different points in time and the system would need the transaction rate as a control. If the transction rate does not meet the expected trading objective of the traders it leads to quote modifications. Our contention is that we can study the trading based on such aggregate behaviour. The intuition is that when a snapshot of the continuous and dynamic trading process is taken encompassing a day or two of trading, the high frequency and price priority nature of the market design ensures that the trading system characteristics dominate over other important criteria such as dividend signals, consumption beliefs etc .

2.2.4 Information Content Structure (ICS)

There is a collective memory of trading systems that we term as 'Information content structure'.

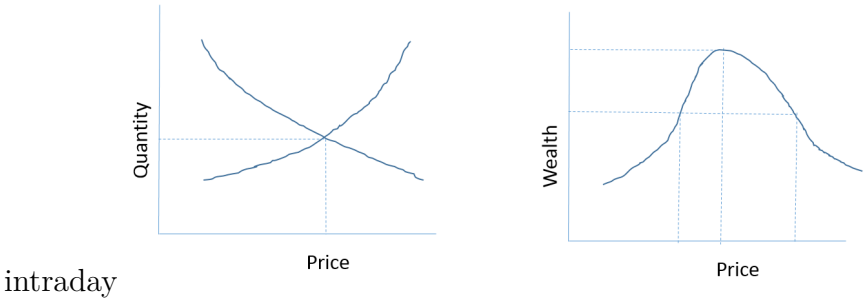

Figure 2: The left panel shows the equilibrium price emerging out of a demand and supply schedule of traders as represented in a total order book that includes latent orders. The right panel illustrates the actual wealth that can be traded at specific prices in the same trading system. The equilibrium denotes the maximum wealth that could be traded. Wealth is the product of quantity and price. (Illustration, not to scale)

The ICS and its relevance in the model is discussed in section 2.3.

2.2.5 From Price - Inventory to Wealth

In the Intraday model we analyse the wealth in the limit order book in an intraday setting and is limited in scope when compared to models traditionally used in market microstructure. For example, in the Smooth Trading Kyle et al. (2018) model disagreement about the precision of private signals motivates trades. Smooth trading analyses: Inventories, Liquidity, Prices, Dividends. We assume that any dividend is neither paid nor announced in the time period of our analysis. A demand and supply dynamics can easily translate into a wealth time dynamics as given in fig 2.

2.2.6 Market clearing and transaction rate or rate of trades are concepts of business time

In the context of continuous auctions, market clearing is a weaker notion as it does not involve time. Since, market clearing involves both quantity and price, it can be conceptualised in terms of wealth used with transaction rates which is a market observable. Fig. 3 shows the emergence of a trade at an equilibrium price from the wealth offered and

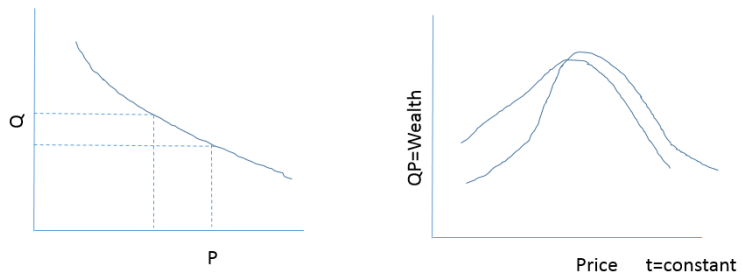

Figure 3: The right panel shows the wealth offered and bid in the quotes at different prices at a particular instant in the LOB. (Illustrations not to scale)

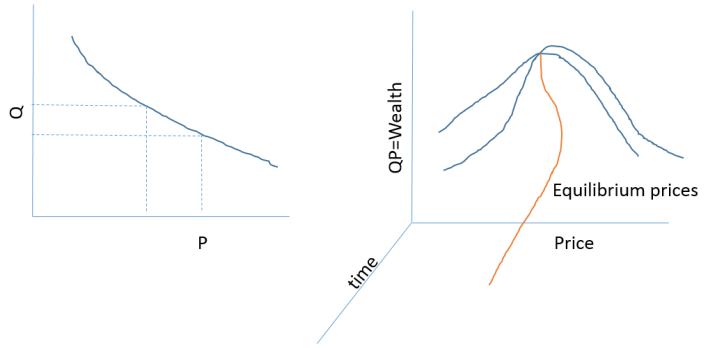

Figure 4: The plot in orange gives the wealth traded at equilibrium price over a period of time in the limit order market (Illustrations not to scale)

bid.

Introducing a time axis we can plot all the auctions. In fig. 4 the plot gives the wealth traded at equilibrium price.

Fig. 5 shows that while equilibrium trading is expected, trading is more likely to follow non-equilibrium path largely due to the lack of time for the information flow required for the equilibrium. In high frequency markets, market clearing will not be met transaction to transaction. It can only be established in coarser time periods. To build predictability into trading, traders would want to see the rate of wealth traded to be constant. If the rate of wealth traded is not constant, they need to adapt their trading strategy to the prevalent scenario. Thus, $\frac{d}{dt}QP = constant$. However, such a wealth transaction rate is specific to the trader. Hence, there will be heterogenous beliefs in $\frac{d}{dt}QP$. Traders would want accumulation or liquidation of stocks leading to constant change in wealth in their

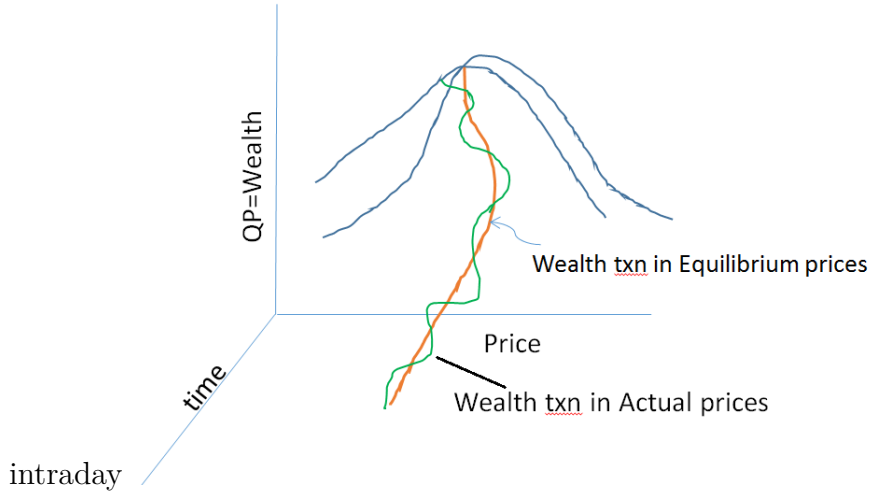

Figure 5: When trading in disequilibrium, the path of trading fluctuates around the equilibrium path

own books. An observer of the trading system would however see that the system will try to go towards optimisation of the wealth transacted by controlling the trading rate.

2.2.7 Optimal rate of transactions

What is the optimal rate of transactions at which the market can clear? Since transactions take place in disequilibrium the objective of the trading system will be to minimise the error from the equilibrium so that it is as close to the market clearing line as possible. The system will attempt to reduce the variance of the wealth transactions in actual versus equilibrium prices. i.e Min$|\sigma_e^2 - \sigma_x^2|$. Since we do not know the equilibrium trading rate, the system will vary from slow trading rate to faster trading rate. That is from low quantity traded to large quantity traded.

Equilibrium will be transitory in this model given the assumptions. Hence, it is difficult to ascertain the equilibrium prices. Other signals such as dividends and investor's utility come into play only when the time horizon of analysis increases.

2.3 Information Content Structure

What is the role of intraday transaction prices, if we argue in proposition 1, that they do not incorporate information fully. Prices in these markets are not driven by ordinary investors. Ordinary investors refers to investors who do not trade large quantities. They are set by large investors, who today have access to the entire order book dynamics from the exchange. They can monitor the limit orders from the limit order book and infer latent order book through the market orders and trade execution. We represent this information through a structure that accumulates the price shock proposed by a quote over time.

2.3.1 Price, Quantity and Time coupling

The coupling between the price shock e, quantity of bid Q and time T_a, in which the bids or offers are communicated, conveys the private information of any agent that arrives into the market. This coupling, called the Information Content Structure $\theta = fe, Q, T_a$ exists and is defined as the total price shock communicated by the agent in unit time. If there already exists an agreement on the public signal, the information that can come with the quote is the price shock carried through a particular quantity of the asset and communicated through a series of quotes over a given time period. This quantifies the agents assessment of the value of his private information. A simple illustration of the ICS is provided in Appendix B.

To explain our intuition related to the ICS lets evaluate a stochastically evolving ICS with transaction prices as bounds.

2.3.2 Stochastically evolving ICS

Consider a two dimensional plane explaining the snapshot at a particular time of the relationship between supply and demand at different prices of the asset. This is essentially the stacking of buy orders and sell orders (both visible in order book and outside the order book) at the respective prices as per the demand and supply schedules of the traders. We assume that private signal for a trader, e, remains constant in the day. We can replace the price axis with this signal. There are strategic buyers and strategic sellers who have this private signal. If these buyers and sellers remain fixated on this signal, more likely the no-trade equilibrium would emerge in several auctions. Buyers and sellers who do not intend to come to the market (through revision of bids) in the event of such equilibrium can be considered as non-participating in the auction. Those buyers and sellers who come into the market treat the private information as the signal around which they set up their demand / supply schedules. Traders have a need to avoid the risk of non-execution. In the intraday trading model this need is explicitly to avoid the uncertainty related to the new public signal in the next trading day.

The non-strategic traders noise traders $NSNT$ are zero intelligent in our set up. They observe the public signal y_p, but they intend to trade irrespective. Their behaviour would be, firstly, not to incur losses in the trading day, and secondly move towards y_p. Their instantaneous behavior is driven by the best bid and offer. They build a demand and supply schedule around the current best bid and offer with the overall design to move towards the public signal. This implies they are indifferent to trading small quantities at the best bid or best offer in the direction of y_p. Given their need to make a non-negative trade, if they move further away from y_p, it implies an increase in the expectation of a change to the public signal . Such a visualisation is important from the perspective of considering search as the tatonnement mechanism as proposed in Pani (2020).

Let $I_t, I_0 = 0$ represent the random variable ICS at time t which is the market's cumulative

volume-weighted price shocks. Divide the trading day $0 \leq t < T$ into $\mathcal{P}$ equal partitions of length $\triangle t$. We want $\triangle t$ to be relatively coarser so that it includes a few transactions. Each partition would have held a number of auctions (number of auctions is not constant). The winning bids in these auctions will define a price range in terms of the $e^{max}$and e^{min}. If we sum quantity-weighted price for each quote whose price shock lies within this range of signal, we get the update to the ICS. ICS over the trading day $t = T$,having $\mathcal{P}$ and j quotes in each partition, gives:

$$I_T = \sum_{i=1}^{\mathcal{P}} \sum_{j=1}^{j} e_{ij} Q_{ij} \psi_{ij} \tag{2}$$

where $\psi_{ij} = 1, \forall a \leq e_{ij} \leq b$ else $\psi_{ij} = 0$ AND

$a = e_i^{min}$ if $e_i^{min} < 0$ else $a = 0$ AND

$b = e_i^{max}$ if $e_i^{max} > 0$ else $b = 0$

The intution is that in any small time interval we may have transactions where signed shocks could be all positive, all negative or mixed. In the first two cases any positive or any negative shock could also have been a transaction price and in the third case the range is from maximum and minimum. These quotes have participated in the auctions and had a higher chance of converting a trade over quotes that lie outside this bound. Based on past experience, the traders infer from the datapoints related to uncleared quantities, available and unavailable prices. In the subsequent pre-opening auction or mid-day price auction, these aspects of the market microstructure apart from fresh news and information needs to be incorporated into the public signal.

2.4 Information Aggregation Into The Price System

The public signal of an asset, y_p is fully informative. (Refer Appendix A for proof) The price system of assets in such markets is the set $\{Y, y_p\}$. (Refer Appendix A for proof)

We invoke the accepted but underrated result from Easley and O'hara (1992), that a period of no trade also conveys as much information to the market as a period of trades. We interpret this to imply that the price systems that we want to examine are unique, informationally more connected and has sufficient number of participants who are actively engaged in either buying or selling the asset. Passive investors who occasionally come into the market and observe the market from a distance would try to extract information from the prices. Active investors on the other hand will go beyond the prices. In these markets both trades and lack of trades signal information.

We develop a hypothesis that there exists a notional system with the prices represented by the public signal of the asset prices around which the regular participants in the market trade. The public signal is updated at a low frequency. Many equity markets employ a pre-auction to arrive at the public signal before the auctions could start in the day. Some markets, for example Borsa Istanbul (Turkey) deploy such a price discovery preauction at the beginning of the day and also mid-day. In markets that do not employ a pre-auction, early trade serves the same purpose. The pre-auction usually employs a price priority instead of a price-time priority. The series obtained from the public signal is a subset of the price system of assets in such markets; The set $\{y_p\}$ records changes at a low frequency. The second moment of the returns from this series gives the measure of information aggregated in the markets. Uncertainty regarding the source of the update to the public signal is necessary in such markets. In a typical trading day, only a fraction of the asset owners come into the market to sell and only a fraction of potential buyers come into the market. The stochastic evolution dissociates a deterministic association of the private information to the fortunes of the larger ecosystem. Essentially it means

that if the market continues to be liquid, private information can result in larger price shocks only through larger exchange of quantity of goods and passage of time. Such an interpretation is trivial.

3 The Model

Consider a market in which a risky asset (equity stock) is traded in exchange for a monetary holding. We envisage a model in which trade is motivated by a need to change the state of holding i.e the traders with risky asset look to convert the asset into the lower risk monetary holding and vice versa.

The utility function for such a system is assumed to take the form in (3):

$$U\pi_c s = -e^{-A\pi_c s} \tag{3}$$

Here π_c is the wealth that is converted or transformed or asset that has changed state. In a time interval dt the wealth that is converted is the transaction rate $\lambda * s * Y$, where Yt is the price of the asset bought or sold.

Let Xt denote the wealth in the visible limit order book. We first set two timescales. A high frequency scale (t) with resolution that can capture the time involved in characteristic quoting activity. And second, a low frequency time scale (T) where the resolution can capture the open auctions that discover the public signal. This intraday trading model straddles a single increment in this time scale. It begins with $T = 0$ and ends at $T = 1$. The events begin after the first pre-auction that discovers the public signal of the Asset price denoted by y_p^0 and ends after the next preauction y_p^1. The trading activity for the

day would have ended between these two events. The asset is always valued at the last transaction price.

3.1 Wealth in the LOB

Let St denote the inventory of the asset in the limit order book (LOB) and Mt denote the monetary cash holdings. The wealth at the beginning of the day is,

$X^0 = ST = 0 * y_p^0 \ MT = 0$ and wealth at the end is $X^1 = ST = 1 * y_p^1 \ MT = 1$. During the day the wealth in the LOB is $X_t = St * Y_t \ Mt$. There is addition and deletion of inventory and monetary holdings through the process of quoting, transaction (state conversion & subsequent withdrawal) and cancellation of quotes.

$$dXt = X_t - rdt \ \ \phi dt - \phi C_D dt - \delta dt \ \ w_t dt - C_D w_t dt - \pi_c dt - \pi_c C_T dt \ \ \sigma_W dW \tag{4}$$

Where the notation denotes the following:

ϕ : The fraction of the wealth that is reinvested or redeployed immediately after conversion. This comes from the non strategic noise traders.

π_c : Converted or wealth whose state was already transformed. This is the control in our model.

C_D, C_T :C_D is the transaction cost incurred per unit of wealth during deployement and C_T is the transaction cost incurred during the transaction.

δ is quote cancellation. Some of these may return through quote modification but is captured in the parameter w_t.

w_t : Wealth getting added through quotes coming from latent to visible LOB.

(4) can be re-written in a simplified form by considering the total cost on the system other than state conversion as:

$Ct = r \;\; \phi C_D \;\; w_t C_D$

Cost= Transaction cost + Time value of money + risk of non-execution.

Time value of money and risk of non-execution influence the quoting behaviour - the behaviour of cancellation and modification at trader level. Cost is one of the drivers of any trading process. From the perspective of the trader, the cost is reflected in the tatonnement process and hence indirectly into the trading system. We are interested in the perspective of the system.

The assumption is that while individual traders react to the slower or faster transaction rate that affects their time value of money or risk of non-execution, it can have a system level effect only after a certain threshold is crossed.. The net addition of wealth factor in terms of inventory being offered or bid as It, combining parameters ϕ, w_t. Further, we state the conversion as $\lambda * s$, where λ is the trading intensity and s is the averaging trade size that is assumed constant for the system. Thus the system controls through the change in the trading intensity.

$$dXt = -CX_t dt \;\; ItY - \lambda * s * Y1 \;\; C_T \;\; \sigma_W dW \tag{5}$$

Let X^c denote the wealth whose state has been converted and X^R denote the wealth that awaits conversion. This wealth will have exposure to the risk emerging from the public signal if the conversion does not take place during the trading hours. In (5) note that the valuation of π_c is based on the transaction price Y. X^R evolves during the day and once the new public signal is ascertained, it transforms as X^1. X_t and X^R seem to be

measuring the same quantity, except that the cost in the system is incorporated into the former. X^R transforms into X^1 at the end of the period after incorporating the ICS and the model assumption (restriction) to keep the non-strategic noise traders wealth neutral. Hence, the discrete process $y_p^0 \rhd y_p^1$ incorporates market memory (ICS) as well as adjusts to incorporate the trading costs and losses of non-strategic noise traders.

$$X^c = {}_t^T \pi_c s ds \tag{6}$$

$$X^R = X^0 \ {}_t^T dX \tag{7}$$

(7) can be better modelled using the process in (8) where λ_t is the trading intensity or rate of trading

$$dX^R = dX - \lambda_t * dY \tag{8}$$

In this article, we envisage a model in which the purpose of the system is state conversion. The system incurs a cost that includes time value of money, transaction costs.

Our problem is to maximise the utility of the trading system.

$$V = J\pi = \max_{\pi s} E \left\{ {}_{s=t}^{\infty} -e^{-A\pi_c s} \right\} = \max_{\pi s} E \left\{ {}_{s=t}^{T} -e^{-A\pi_c s} \ rX^1 \right\} \tag{9}$$

Note that the wealth in the limit order book is only a small fraction of the wealth that can be represented by the asset (potentially). The pre-auction or initial auction in the

day sets the wealth X^0, X^1. Change in wealth is through an inflow of wealth into the limit order book, cancellation and state conversion. The asset that has converted its state can come into the market again later. We interpret (9) in the following way: Since the purpose of the trading system is state conversion, its utility for a single day of trading is given by the first term. And the second term gives the utility for the future trading days. Non optimal trading on any day shall impact $X^1$by the change in the public signal of price or the inflow and cancellations in the subsequent periods. (9) is subject to the followng constraints.

Wealth constraint defined by the process dXt given in (4). The asset price Y defined by the process (10) and public signal defined by the process (11).

$$dY = \alpha_1 Y dt \;\; \sigma_Y dB_1 \tag{10}$$

$$dy_p = \alpha_2 y_p dt \;\; \sigma_p dB_2 \tag{11}$$

(11) is assumed to be an approximation of the signal observed due to the discrete process that gets generated basis the market memory (ICS). i.e $y_p^0 \rhd y_p^1$ conditioned on the ICS. In the current model ICS becomes zero after the pre-auction adjustment while the public signal in (11) is a continuous time adjustment.

Lets reconcile with our two kinds of traders. Strategic traders observe the asset price Y and Non-strategic traders observe the public signal y_p. Strategic traders are believed to posses private information, while non-strategic traders do not possess private information and trade on the precision of the public signal. The non-strategic traders who trade repeatedly and continuously in the market are termed non-strategic noise traders. Their

asset holdings undergo the state transformation several times. In the absence of market makers, such noise traders play an important role. We visualise that, for such markets to be sustainable, these noise traders should not make losses or bear the transaction costs. We do not explicitly model this constraint for two reasons. Firstly, the system is an open system. Wealth flows from the limit order book to the total order book and vice-versa. Secondly, we assume the presence of spreads and adjustment to the public signal would incorporate this requirement as the noise traders trade around the public signal and should influence the same. Both strategic traders and nonstrategic traders may use limit orders or market orders

We conjecture and verify a value function of the form VX, y_p, Y.

$$VX, y_p, Y = -exp\psi_0 \ \ \psi_w X \ \ \psi_p y_p \ \ \psi_Y Y \tag{12}$$

The corresponding HBJ equation becomes

$$\begin{aligned}
&0 = \\
&\max_{\pi_c} \left\{ \frac{-e^{-A\pi_c} \ \ rX^1}{V} \ \ \psi_w - CX_t \ \ ItY - \lambda sY1 \ \ C_T \right\} \\
&\psi_Y \alpha_1 Y \ \ \psi_p \alpha_2 y_p - \frac{1}{2}\sigma_w^2 \psi_w C \ \ \frac{1}{2}\sigma_w^2 \psi_w^2 - CX_t \ \ I_t Y - \lambda sY1 \ \ {C_T}^2 \\
&\frac{1}{2}\sigma_Y^2 \psi_Y \alpha_1 \ \ \frac{1}{2}\sigma_Y^2 \psi_Y^2 \alpha_1^2 Y^2 \ \ \frac{1}{2}\sigma_p^2 \psi_p \alpha_2 \ \ \frac{1}{2}\sigma_p^2 \psi_p^2 \alpha_2^2 y_p^2 \\
&\alpha_1 Y \psi_Y \psi_w - CX_t \ \ I_t Y - \lambda sY1 \ \ C_T \\
&\qquad\qquad \alpha_1 Y \psi_w^2 - CX_t \ \ I_t Y - \lambda sY1 \ \ C_T \ \ \psi_w I_t - \lambda s1 \ \ C_T
\end{aligned} \tag{13}$$

The optimal control for above HBJ equations is obtained as:

$$\pi_c^* = \frac{-1}{A} log \frac{\psi_w 1 \ \ C_T V}{A} \tag{14}$$

We put back π_c^* into (13) and expand the squares to obtain the following,

$$
\begin{aligned}
&0 = \\
&-\frac{\psi_w 1\ C_T}{A}\ \frac{rX^1}{V}\ \psi_w - CX_t\ I_t Y - \lambda s Y 1\ C_T \\
&\psi_Y \alpha_1 Y\ \psi_p \alpha_2 y_p - \frac{1}{2}\sigma_w^2 \psi_w C \\
&\frac{1}{2}\sigma_w^2 \psi_w^2 C^2 X_t^2\ 1\ \lambda^2 s^2 Y^2 1\ C_T{}^2\ 2\lambda s Y 1\ C_T \\
&-2CXY1 - \lambda s Y 1\ C_T \\
&\frac{1}{2}\sigma_Y^2 \psi_Y \alpha_1\ \frac{1}{2}\sigma_Y^2 \psi_Y^2 \alpha_1^2 Y^2\ \frac{1}{2}\sigma_p^2 \psi_p \alpha_2 \\
&\frac{1}{2}\sigma_p^2 \psi_p^2 \alpha_2^2 y_p^2\ \alpha_1 Y \psi_Y \psi_w - CX_t\ I_t Y - \lambda s Y 1\ C_T \\
&\qquad \alpha_1 Y \psi_w^2 - CX_t\ I_t Y - \lambda s Y 1\ C_T\ \psi_w I_t - \lambda s 1\ C_T
\end{aligned}
\tag{15}
$$

We separate the coefficients of the variables $X_t, X_t^2, Y, Y^2, y_p, y_p^2$ and equate to 0. Equating the constant term to 0 yields:

ψ_0

$$
\begin{aligned}
&-\frac{\psi_w 1\ C_T}{A}\ \frac{rX^1}{V} - \frac{1}{2}\sigma_w^2 \psi_w C\ \frac{1}{2}\sigma_w^2 \psi_w^2 \\
&\qquad \frac{1}{2}\sigma_Y^2 \psi_Y \alpha_1\ \frac{1}{2}\sigma_p^2 \psi_p \alpha_2\ \psi_w I_t - \psi_w \lambda s 1\ C_T = 0
\end{aligned}
\tag{16}
$$

X

$$
-C\psi_w = 0 \tag{17}
$$

X^2

$$
\frac{1}{2}\sigma_w^2 \psi_w^2 C^2 = 0 \tag{18}
$$

Y

$$\psi_w I_t - \psi_w \lambda s1 \;\; C_T \;\; \psi_Y \alpha_1 \;\; \sigma_w^2 \psi_w^2 \lambda s1 \;\; C = 0 \tag{19}$$

Y^2

$$\frac{1}{2}\sigma_w^2 \psi_w^2 \lambda^2 s^2 1 \;\; C^2 \;\; \frac{1}{2}\sigma_Y^2 \psi_Y^2 \alpha_1^2 \;\; \alpha_1 \psi_Y \psi_w I_t$$
$$- \alpha_1 \psi_Y \psi_w \lambda s1 \;\; C_T \;\; \alpha_1 \psi_w^2 I_t - \alpha_1 \psi_w^2 \lambda s1 \;\; C_T = 0 \tag{20}$$

y_p

$$\psi_p \alpha_2 = 0 \tag{21}$$

y_p^2

$$\frac{1}{2}\sigma_p^2 \psi_p^2 \alpha_2^2 = 0 \tag{22}$$

XY

$$-\sigma_w^2 \psi_w^2 C - \alpha_1 \psi_Y \psi_w C - \alpha_1 \psi_w^2 C = 0 \tag{23}$$

XY^2

$$\sigma_w^2 \psi_w^2 C \lambda s1 \;\; C_T = 0 \tag{24}$$

The above system of equations 16-24 can be solved numerically to obtain the results for

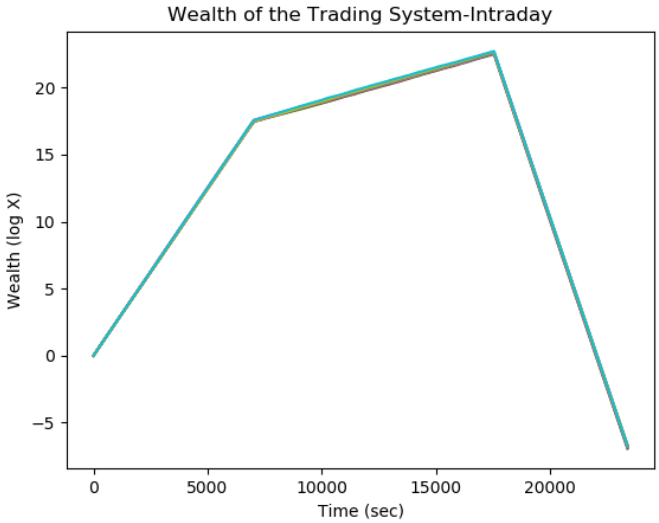

Figure 6: Wealth (log X) of the trading system as it evolves through the trading day 9.30 am to 4.00 pm as per (Equation 2.8). The values of the constants C = 7.69231e−7, C_t = 1.19e−9, σ_W = 6.5372e−4, σ_Y = 3.2686e−4 , σ_P = 6.5372e−5, α_1 = 4.27e−7, α_2 = 4.274e−5, λ = 4,s = 100, n_r = 1.0025, n_r = 1.0005 and n_r = 0.995 in the first 30 percent of the trading period, 30-75 percent and the remaining trading period respectively. The plot shows 10 simulations and it remains identical even for a large number (>1000) of simulations.

the unknowns $\psi_0, \psi_w, \psi_Y, \psi_{y_p}$. We are not aware of any closed form solutions to the above equations. While selecting the solutions (14) serves an additional constraint due to which ψ_w is non-negative. Fig. 6 plots the wealth in such a trading system during a typical day. Additionally for the simulation we have assumed a specific relationship between the trading rate and quoting rate given by n_r. The intuition is that in the first period there is a high level of quoting resulting into the increase in wealth. In the second period, the rate of quoting activity more or less tracks the trading rate. In the third period the quote modifications and cancellations come in along with higher trading rate. The system's wealth ideally should end in negative showing the cost of transaction which in this model is borne by long term strategic investors.

In Fig. 7 the sample path of intraday prices is provided. This is governed by the SDE (10). Fig. 8 gives the trading intensity with different precision of the signals. In optimal trading the trading intensity shall remain stable if we assume an average trade size (for example 100 shares per trade). In our model the precision of the longer term signal is

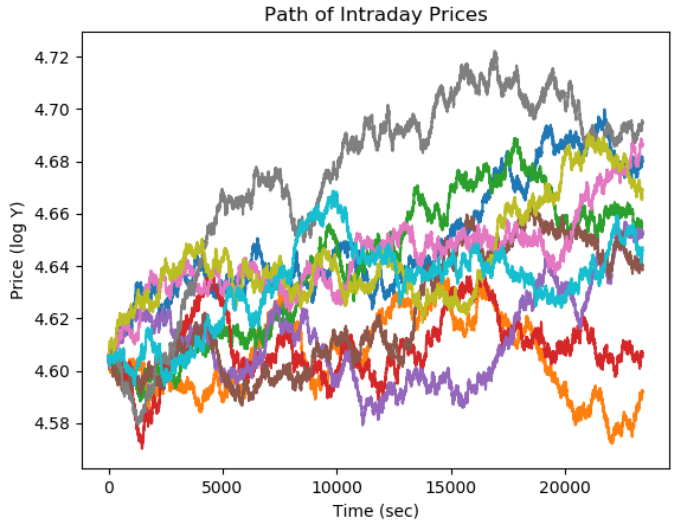

Figure 7: Sample path of intraday log-prices as per (13). The values of the exogeneous constants $C = 7.69231e{-}7$, $C_t = 1.19e{-}9$, $\sigma_W = 6.5372e{-}4$, $\sigma_Y = 3.2686e{-}4$, $\sigma_P = 6.5372e{-}5$, $\alpha_1 = 4.27e{-}7$, $\alpha_2 = 4.274e{-}5$, $\lambda = 4$, $s = 100$, $n_r = 1.0025$, $n_r = 1.0005$ and $n_r = 0.995$ in the first 30 percent of the trading period, 30-75 percent and the remaining trading period respectively. The plot shows 10 simulations and it remains identical even for a large number (> 1000) of simulations. The initial value of the asset price assumed is $log100$.

a constant in intraday and changes before the next trading day. When the precision of the signal from the intraday prices is low (i.e the volatility or variance is high) it results in lower trading intensity. Only when the precision of the intraday price signal is high, relatively the trading intensity is higher. One way the trading system can increase the trading intensity is by increasing the precision of the intraday trading prices. In a period where the trading intensity is not satisfactory to the traders, it may call for modification of quotes to reduce the variance (increase the precision of the signal).

In Fig. 9, we describe the relationship between the trading activity and the quoting activity in terms of the inventory I_t. Intuitively, the highest rate of trading can be achieved in an hypothetical system where incoming market orders liquidate the exisiting limit orders. This is depicted in the lower values of n_r. However, every trading system will settle down to its characteristic trading intensity with a diminishing influence of the quoting multiple.

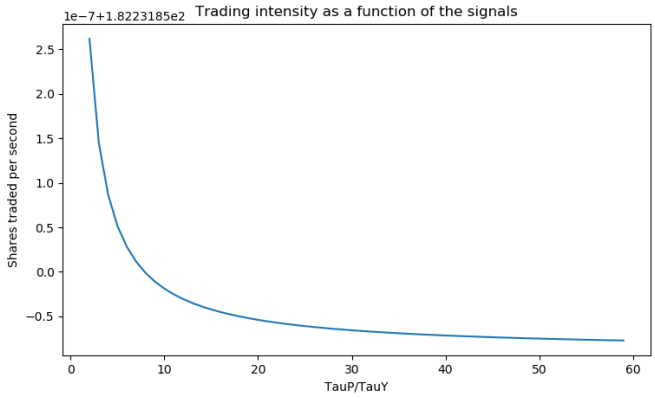

Figure 8: Shares traded in optimal trading with the precision of the signals. The long term signal remains constant through out the day in our model. The values of the exogeneous constants $C = 7.69231e{-}7$, $C_t = 1.19e{-}9$, $\sigma_W = 6.5372e{-}4$, $\sigma_Y = 3.2686e{-}4$, $\sigma_P = 6.5372e{-}5$, $\alpha_1 = 4.27e{-}7$, $\alpha_2 = 4.274e{-}5$, $\lambda = 4$,$s = 100$, $n_r = 1.0025$, $n_r = 1.0005$ and $n_r = 0.995$ in the first 30 percent of the trading period, 30-75 percent and the remaining trading period respectively. In the x-axis, $\tau_P = 1\sigma_P^2$ is the precision of the public signal which is high and does not change during the trading day and $\tau_Y = 1\sigma_Y^2$ is the precision of the intraday trading prices relatively lower.

4 Microstructure and Trading effects

In Appendix 3, using a few illustrative examples we explain how the microstructure and trading can distort the information that get incorporated into the ICS, reinforcing the proposition in this article that intraday prices cannot be used to aggregate information and only the public signal, that emerges out of a larger consensus and lower frequency, aggregates information. The lower frequency of aggregation cancels microstructure effects.

5 Concluding Remarks

The theoretical conceptualisation of markets as a system has been accepted since the eighteenth century and is one of the few steady constructs through the different ideas that have emerged during this period. When markets are viewed in high resolution, in order to specifically appreciate the enhanced role of market microstructure variables, we qualify markets as a Trading System. The understanding of a mechanism is adapted to

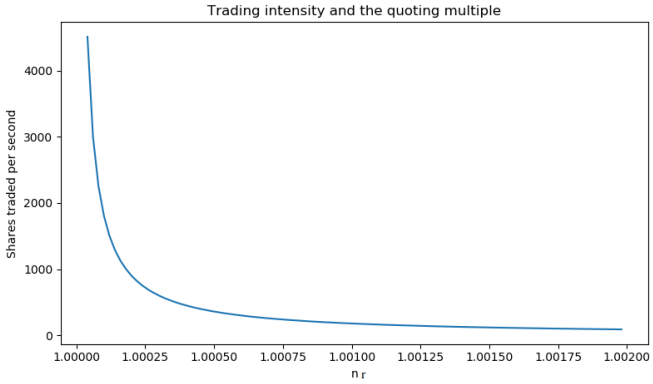

Figure 9: n_r describes the relationship between the net addition of wealth in terms of inventory (quoting activity $I_t = n_r * \lambda * s$) and the traded assets (y-axis) $\lambda * s$. n_r is the multiple. The values of the exogeneous constants $C = 7.69231e{-}7$, $C_t = 1.19e{-}9$, $\sigma_W = 6.5372e{-}4$, $\sigma_Y = 3.2686e{-}4$, $\sigma_P = 6.5372e{-}5$, $\alpha_1 = 4.27e{-}7$, $\alpha_2 = 4.274e{-}5$.

the trading system view. The current paradigm of a mechanism view taken along with the game-theoretic approach is assumed to be constrained due to fundamental assumptions while working with these tools. A few such assumtions are relaxed. The new framework continues to work under the Information processing role of modern markets. The output variables are at the aggregate level and can be fewer in number (One in the current paper - the trading rate). This is simply an acknowledgement of limitations of communications in high frequency and when the analysis is at high resolution. There is an emphasis of the role of markets as places of exchange. This role is of primary importance if not more than price discovery or information assimilation.

The apporach used in this paper, is only one of the several possible to achieve the above objectives. In summary, the information content of orders and intraday trading in limit order markets is examined under imperfect competition. Such a setting is motivated by a model without a designated market maker, a design closer to reality in modern markets. Further, prices need not emerge out of an equilibrium. Trading is motivated by a need to change the state of the asset held with the participants. The rate of such state transformation is characteristic to the trading system as it relates to the expected outcome during the trading day. Any unexpected deviation from characteristic trading rates results into quoting activity that can counter the deviation. Traders with private

information use the emerging asset price as the signal and traders without private signal trade on the public signal. During intraday trading, a representation is proposed where information with market participants is represented in an interim Information Content Structure (ICS). The ICS serves as the market's memory of the events during the trading day and is the cumulative volume-weighted price shock per unit time incorporated by the trader's quotes. Information is aggregated to the public signal of the asset price which is deemed to be a constituent of the price system. A low frequency aggregation based on larger consensus into the public signal cancels microstructure and trading effects.

Acknowledgments

The would like to thank Albert Pete Kyle for comments received on an early version of this article which helped improve the contents in this manuscript. The author would also like to thank Susan Thomas, Rituparna Sen, Ranjan Chakravarty, Ramesh Bhat, Chandan Dasgupta, Mayank Joshipura and participants at the World Finance Conference, Norway, August 3-6, 2021 for their helpful comments and suggestions.

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A Proofs

Proof of proposition 1

Example: To understand the proposition consider the case where there are 2 owners of the asset, holding n_1 and $n - n_1$ quantity of the asset respectively. Let the last transaction price be Y_{lp}. The actual wealth and wealth-public value of the two owners is $n_1 * Y_{lp}$, $n_1 * y_p^0$ and $n - n_1 * Y_{lp}$, $n - n_1 * y_p^0$. If the first owners' holding is sold at price Y_{11} when the public signal is y_p^0, we want to show that Y_{11} is not fully informative if the second owner has not participated in the auction. Here the subscript of the price shows the number of

the auction and identifies the owner. Thus, Y_{12}, signifies the price of first auction-second owner.

Had the second owner participated, it indicates that the asking price of the second owner Y_{12} is greater than Y_{11}. If Y_{11} is fully informative, it incorporates all the available public and private information. Thus Y_{11} is the new public signal. The choice for the second owner is to wait for the asking price (Y_{12}) to arrive or sell at a discount to his initial asking price. Other traders too learn about the transaction price and prior expectation of an important owner of the asset. The buyer in the first transaction will try to liquidate before the ask price of the second owner, $Y_{11} < Y_{21} < Y_{12}$. Given the uncertainty over future news flow or private information, the second owners optimal choice may be to liquidate at Y_{11}. More so because, if $Y_{11}$is the public signal, only strategic buyers are present above this price as buyers.

An owner of a security may not participate in an auction on a given day due to several reasons: portfolio decisions such as income needs, holding period, taxation etc, he may want to come in at specific prices with market orders or limit orders, he does not have a demand or supply schedule around the prices in which market is currently trading. When, the second owner has not participated, he still faces a change in wealth due to the transaction. The new values being respectively $n - n_1 * Y_1$, $n - n_1 * Y_1$. If $Y_1 < Y_{lp}$ he was better off being in the market and participating in the first auction as he could avoid the uncertainty over liquidation price. If $Y_1 > Y_{lp}$, while his actual wealth has increased, his expected wealth based on fundamental value too has changed. $Y_1 < y_{p1}$ is more likely given that non strategic buyers are likely to be present in this scenario at Y_1. $Y_1 > y_{p1}$ is good outcome for the second owner as it reveals the purchase in the first auction by a strategic buyer. But it also means that there will be no buyers above Y_1 and we get a no trade equilibrium. So the second owner has to liquidate below Y_1 whenever he liquidates. He was better off participating in the first auction and selling at $Y_1 - \delta$.

Extension:

Extending the analysis to a large number of asset owners $O_1, O_2..., O_m 1 \leq m \leq n$. A fraction of the asset owners may actually participate in the auction. An owner of a security may not participate in an auction on a given day due to several reasons: portfolio decisions such as income needs, holding period, taxation etc, he may want to come in at specific prices with market orders or limit orders, he does not have a demand or supply schedule around the prices in which market is currently trading. A non-participating owner still faces a change in wealth due to the transaction and an uncertainty over future liquidation prices. If intraday transaction prices were fully informative, a trading day represents a continuous stream of transaction prices resulting in continuously fluctuating public signal of the asset price, thereby rendering the assumption of the update to the public signal at the end of the trading day meaningless. Just as the analysis for two owner case, for each of the transaction prices, non winning bids / offers simply reveal their strategies and non-participating traders would be better off participating than facing unending and continuous uncertainty if the transaction prices are considered to be fully informative.

Proof of proposition 3

The proof is continuation of the proof of proposition 1. If public signal was not fully informative, then the non-strategic traders will make an adjustment to the public signal for their trades with the help of value judgments regarding the impreciseness. The signal will cease to have the characteristics of the public signal. The public signal is updated from the ICS at the end of the trading day.

Proof of proposition 4

From proposition 1, if the prices were the only aggregator of information in such markets, then there would be multiple signals visible to the traders. For example, EY, Y_{max}, Y_{min}, $Y_{avg}$and others. Different traders would use different signals with a knowledge that it

is an incomplete signal. As a result the no-trade equilibrium would increase. Hence, a common signal of the prices, such as the public signal y_p (from proposition 3) needs to be present in the price system. However, traders who come into the market infrequently and observe the market from a distance will still refer to the prices.

B Example: Illustration of ICS

The agent's intent is to communicate a price shock of e through a specific quantity Q $Q < n$ and in a time interval T_a conditioned by his private information. The actual information content imparted to the market per unit time by the jth $j \in N$ agent may be described as:

$$\theta_j e, Q, T_a = {}_{i=1}^{i} \frac{e_i * Q_i}{T_i} \tag{25}$$

T_i is the time spent by the agent in each auction, ${}_{i=1}^{i} T_i = T_a$ and Q_i is the quantity bid or offered and $|e_i|$ is the price shock per share offered in each quote. Equation (2) is the total price shock in unit time communicated by the jth agent to the market. If there are k trades during the trading day, each trade is the conclusion of an auction. Each auction will have a number of agents transferring information into the market. So the total information transferred in the kth auction is ${}_{j=1}^{j} \theta_j * \phi_j$, where ϕ_j is the binary information whether the agent had participated in the auction or not. The total information offered during the trading day that had k auctions in which a maximum of j agents could have participated is given by ${}_{k=1}^{k} {}_{j=1}^{j} \theta_{jk} \phi_{jk}$

The total information offered during the trading day is given in (3), where ψ is a check

for genuine and valid quotes:

$$\theta = {}_{k=1}^{k} {}_{j=1}^{j} {}_{i=1}^{i} \frac{e_i Q_i}{t_i} \phi_{jk} \psi_{jk} \tag{26}$$

Such a check is essentially to confirm if the quote is a part of the expanded filtration as operationalised in Chakravarty and Pani (2018). The information content in the winning bids comes from the trade price series. For the k trades, the total information content change is given by θ_a, $\pi_a = {}_{k=1}^{k} \theta_k$, where $\theta_k e, Q, t = \frac{e_k Q_k}{t_k}$. e_k is the price shock incorporated in the trade, Q_k is the quantity traded and t_k is the inter-trade time known as trade duration. If the trade was executed at multiple prices, $\theta_k e, Q, t = {}_{p=1}^{p} \frac{e_{kp} Q_{kp}}{t_k}$. Since markets are able to incorporate only a fraction of the total private information into the information structure of the trades, there is an efficiency of the auction, η, given as:

$$\eta = 2\frac{\theta_a}{\theta} \tag{27}$$

The multiple 2 signifies two bids participating in each trade. If the market is successful to clear the quantities or incorporate the prices an Information equilibrium can exist, $2\pi_a = \pi$ as the efficiency is 1.

However, this is only a theoretical equilibrium as several bids are canceled or revised before they are traded. Further, we have viewed the private assessment of the value of trader's assessment. It is not difficult for the lines between a rationale assessment and a speculative one to get blurred. Hence, all the shock that comes into the market may not carry information and may only represent noise.

One way to correct the analysis is to use the transaction prices as bounds in a stochastically evolving ICS. The ICS is a quantity that accumulates the price-quantity shocks coming

into the market. The shock is not revised if the quote is canceled without a trade or revised. A revision is treated as a new shock as it comes in at a different time and auction. Thus the nature envisaged for ICS is to represent a process of active monitoring of the market by traders. For example, if a trader comes with a price shock e_1, Q_1, t_1 but his trade does not go through for a certain time period, he observes that the price has changed. So he revises his quote to e_2, Q_2, t_2. The market monitoring process ICS does not revise e_1, Q_1, t_1 to e_2, Q_2, t_2. Rather it will have both the information. A revision may have been useful if we wanted to compute, say a second moment to represent the ICS. That is not the role of ICS. The motivation for such a representation of market monitoring pricess is due to the intuition in Easley and O'hara (1992). It is this cumulative knowledge of the market that the market adjusts the transactionintraday prices.

C Illustrative examples: Microstructure, Trading and ICS

Fig. 10 gives the price grid at the start of the day in terms of the price shock e . This price shock that comes with a quote is based on (1) relative to the public signal y_p and remains constant. However, the relative role of each quote may also be viewed with reference to the last transaction price. Fig. 11 gives the relative price grid at time t when the last transaction price is denoted by y_{lp}. The position of y_p is shown in terms of the absolute price grid. In summary, when we refer to the price shocks we refer to an absolute price scale with reference to y_p which has zero price shock. While when we refer to the last transaction price and the relative position of the best bid BB, best offer BO and other price shocks are numbered with reference to the last transaction price.

Large imbalance between buy and sell side volumes

Non-Strategic traders

Consider an auction in progress. Instead of the price, we shall instead use the price shock brought in by the quote (Refer Fig. 10). Assume that the last transaction at e_{lp} has exhausted all buyers and sellers at this price (both orders present in the order book and latent). e_{lp} can be the next winning auction if atleast one buyer and seller come into the market at this price. The next candidate for the winning price is from the range e_{BB} to e_{BO} which also includes e_{lp}. e_{BB} is the highest price bid, irrespective of the quantity bid or time of arrival and is available at the price $e_{lp} - \theta_1$. e_{BO} is the lowest price offer, irrespective of the quantity bid or time of arrival and is available at the price e_{lp} θ_2. $\theta = \theta_1$ θ_2 relates to the cumulative friction that is present in the entire process and has been analysed in depth in market microstructure literature. The zero intelligence behavior of non-strategic traders would be to move towards the public signal of the asset and accordingly choose between e_{BB} and e_{BO}. This leads to further update in the information structure defined by the coupling between price and quantity traded. Such an information content update occurs even if the absolute price represented by the selected e_{BB} and e_{BO} on the price axis defined by e has been achieved earlier as a traded price. Since the market is trading away from the public signal, the quote from the non-strategic trader also gets included into the ICS based on our construction.

Buyers and sellers would not be able to time their presence in the market to perfection and there will be imbalance in volumes. The regular traders understand the usual imbalance. However, if there is a large imbalance (several times higher than daily imbalances) between the quantity offered by the buyers and sellers, there could be two consequences, both involving a price impact in the direction opposite to the excess quantity. If the excess quantity is in the direction opposite to the public signal, it leads to an accelerated movement towards the public signal. For example, the public signal may currently be on

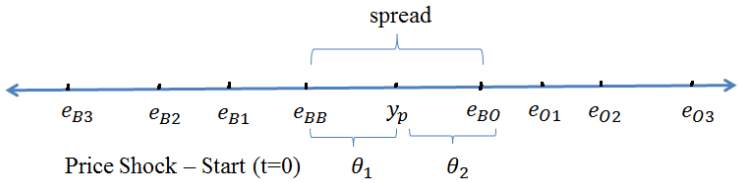

Figure 10: A price grid based on absolute price shock of the quote relative to the public signal at the start of the trading day

the ask side of the prices, while large volume buys would have appeared. On the other hand, if the excess quantity is in the direction of the public signal, it is construed that there is new information that shall update the public signal later. This direction of trade is the only alternative to a no trade equilibrium in a competitive auction. Literally, the excess quantity has to be collected from the opposing camp and the concession has to be made by the camp having the private information as the opposing camp does not have the same information. The price impact is a direct consequence of the information content of the bids and offers in the auction.

Implication of the above description : Large volumes do not come in the quotes of non-strategic traders. Even if the volume is demanded or supplied at the public signal, there may not be enough volume available with the counterparties at that prices. The market cannot absorb the information here without a passage of time and flow of information to counterparties convincing them of the price. Competitive dynamics forces the traders to cut the value of expected gains and pick up volume from the opposing camp. This is a possible explanation to the price impact from large trades. Trade price impact can potentially trap the non-strategic traders away from the public signal in our model due to the constraint of zero-intelligence. They would want to take as little losses as possible, and this can slow down trading intensity.

Strategic Traders

What are the constraints in the price dynamics specific to the strategic trader who comes with the private information. Suppose that a trader comes into the market to sell a large quantity, Q_s, at a price e_θ when the last transaction price is at e_{lp}. He can come in with a single order or a number of orders sequentially as the price evolves. While e_θ can be confirmed to be a positive quantity being a sell order, we are not sure of the sign of e_{lp}. Hence the maximum information possessed by the trader, if it can be incorporated instantly in the market, is either $Q_s * e_\theta\, e_{lp}$ or $Q_s * e_\theta - e_{lp}$. The trader's private information and his assessment of the impact of the private information on the public signal and fundamental value are important to decide his strategic choices. Without incorporating the expected future alternatives, if the weighted average price of his trades has been e_w, the information incorporated from the trader is $Q_s * e_w$. The boundary conditions for him to choose to trade is $e_w > 0$ or some specific minimum price of his choice $e_w \geq e_{min}$. Thus he has received some gain for the private information.

However, if in the trader's assessment, the private information would decrease the fundamental value or public signal in the future by an amount θ_3, his limit moves to $e_w - \theta_3 \geq e_{min}$. He may have to trade at a value unfavorable to this only if similar private information is available with some other traders and hence there is competition to act on the information. In such circumstances, the information incorporated into the traded price, would be higher than what was being offered initially. Thus trade prices can erroneously incorporate trading noise along with information. Note that the seller will ask for a higher price only in case his private information gives a positive reinforcement to the fundamental value and public signal. He is more likely to trade at $e_\theta <= e_{lp}$ in case of a negative reinforcement.

When the seller comes at the price $e_\theta = e_{lp}$ the buyers cannot ascertain the seller's assessment of his private information. Strategic buyers can either wait for reduction of

the offer price or increase their bids acknowledging the concession from the seller as being enough. The buyers will evaluate the private information of the seller (price and quantity) now by bringing in the time factor. Is the seller impatient? Is it a natural trade from a strategic seller or there are more sellers with similar private information? An assessment that the seller is impatient or that there is competition among sellers to accept lower prices indicating negative reinforcement in the private assessment of the sellers, may persuade the buyers to demand further lower prices.

Collapse of Auctions

If a number of large sellers have private information about the true nature and scale of the negative reinforcement regarding the public signal, the public signal gets contaminated, the buyers may withdraw from the market leading to a free fall in price. This signifies a dissociation from the information aggregation function of the price system in speculative asset markets. A similar situation can play out in the opposite direction as well, where private information contaminates public signal leading to speculative buying.

Bid-Ask bounce

We make additional assumptions regarding the starting equilibrium to simplify the game: Let the quantity held by each trader is 1 unit. This implies there are as many traders as there is quantity available to trade in the market. There are equal number of non-strategic buyers and sellers. If the bid-ask spread is small, as is expected in deep markets in regular trading days, the non-strategic buyers are available from the best bid and linearly drop off in numbers reaching zero at one standard deviation of the daily intraday volatility. Similarly, for the non-strategic sellers. The presence of strategic buyers and sellers is a linearly increasing function away from the best bid and best offer. Some of these strategic traders may have deployed their quotes as limit orders, while some of their quotes may

be in the hidden latent book, that will come into the market as market orders or get placed at an opportune time a limit orders. The supply(S) and demand(B) balance can be represented by the density of the quantity ΔQ. S and NS represent strategic and non-strategic traders.

$$f_{\Delta Q}e|y = B_S e \;\; B_{NS}e - S_S e - S_{NS}e \tag{28}$$

The strategic traders who have traded at e_{lp} may or may not come back into the market in the price range of the best bid - offer. If there is no change in private or public information, the continuity in the auction depends on the non-strategic traders who had traded at e_{lp}. The seller has received the cash and the buyer has received the asset at that price. The only option remains for the cash to bid at e_{BO} and the shares to be placed for sale at e_{BB} to avoid the no-trade equilibrium (Refer fig. 11). Let the next trades start at the e_{BO}. Again strategic sellers at e_{BO} move out of immediate action with the cash, the non-strategic sellers have the option to buy at e_{lp} or e_{BB}. If these traders place their new buy orders at e_{lp}, this becomes the new e_{BB1} (here 1 is a counter for the best bid BB) else e_{BB} continues to be the best bid. Incoming market orders will then decide the next transaction price. e_{BB} is the best bid and e_{O1} is the best offer. If the next seller is prepared to place a market order then e_{BB} becomes the transaction and we get the bid-ask bounce, else a limit order may be placed anywhere from $e_{lp}$and above. This kind of transactions result into the famous bid-ask bounce noticed in continuous double auctions. The dynamics depends on the incoming market orders. On a normal trading day an equal probability of buy or sell market orders can be expected and hence the bounce can be seen. This is another noise that keeps adding to the ICS as the shocks may be coming from non-strategic traders who trade around the best available prices towards the public signal and do not have any private information.

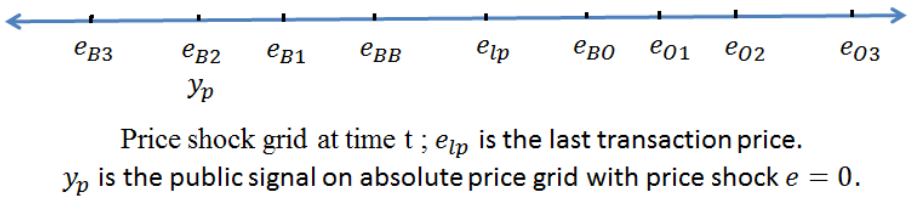

Figure 11: Relative price grid based on relative position of the quote to the last transaction price. The public signal is also shown to refer to the origin of the absolute price grid.

Impatience in trading

Local equilibrium, as above, can reduce the auctions that result into trades. Impatience resulting from non-execution can push the traders from their equilibrium positions even without any new information. Non-strategic traders sense that the trades are likely to be range bound around e_{lp} and hence there will be low volatility from here on during the day's trading. They gather around this signal and commence the trades as given by (10). Information incorporation will remain low given that higher quantum of information is held with the strategic investors. New information may come into the market and can get incorporated into private information by the strategic traders leading to more auctions being won by the strategic traders. In the absence of new information, if any strategic trader becomes impatient (for example, α percent of strategic buyers in (11), especially the traders very close to the trading zone, they may linearly translate their positions by a $\theta/\ -\theta$ increasing the quantity traded, and information incorporation into the ICS is increased in the process due to the revisions.

$$f_{\Delta Q}e|y = B_S e\ \ B_{NS}^{e_{lp}} - S_S e - S_{NS}^{e_{lp}} \tag{29}$$

$$f_{\Delta Q}e|y = \alpha B_S e \;\; \theta \;\; 1 - \alpha B_S e \;\; B_{NS}e - S_S e - S_{NS}e \tag{30}$$